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DEALER BEST PRACTICE SERIES

Managing and Adjusting Major Component Flat Rates

Application | Maintenance | Site Management | **Component Rebuild** | Component Life Management | Safety | MARC Management

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1.0 Introduction

The majority of large mining customers around the world rely on major component exchange to keep their fleets running at high availability. Supporting a major component exchange program requires that component rebuilds be performed on a flat rate basis.

Establishing component rebuild flat rates and keeping them current is a challenge that is important to CRC financial performance. Component rebuild costs change over time due to factors such as parts prices, labor costs, product design updates, age and hours of machines, parts reuse and salvage, and application changes.

This Best Practice discusses an annual work order analysis process to adjust major component rebuild flat rate pricing.

2.0 Best Practice Description

There are a number of factors that determine component rebuild costs. No two component rebuilds cost exactly the same amount. While it is easy to determine an average cost for a specific component, there is always variation from one work order to the next. While establishing a flat rate price, it is important to look at cost variations and manage them as much as possible.

Factors causing rebuild cost variations:

- Parts reusability
  - Total life cycles/ number of prior rebuilds on the component. (Age variable)
  - Maintenance/ fluid cleanliness practices (Customer variable)
  - Severity of application – fuel burn rate (Customer variable)
  - Before versus after failure rebuild contingent damage (Condition variable)
  - Major casting fatigue life (Age variable)
  - Design changes/ product updates (Design variable)
  - Reuse practices – risk avoidance (Shop variable)
- Parts and labor price changes (Price Inflation variables)
- Labor usage
  - Parts salvage requirements (machine shop or outside) (Age/ Condition variable)
  - Before versus after failure. (Condition variable)
  - Work process efficiencies (Shop variable)
  - Use of Reman piece parts (Shop variable)

Note: Excel Pivot Charts or Minitab are useful tools for this analysis.

3.0 Implementation Steps

The process to analyze and adjust component rebuild flat rates involves the following steps;

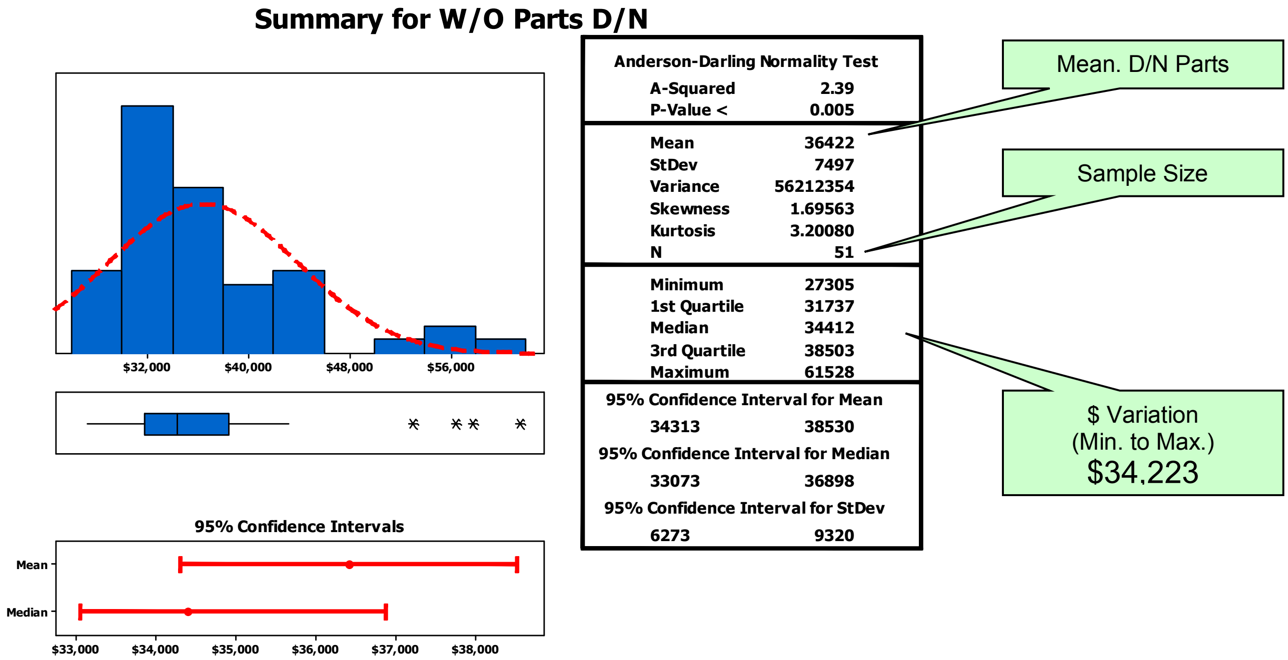
- Collect all recent component rebuild work orders for the component being reviewed.
- Determine the average and median component rebuild costs.
- Measure the variation in the data
  - By customer (maintenance and application factors)
  - By component age (number of prior rebuilds)
  - By component condition (before or after failure)
  - By product updates (design changes)
- Determine reasons for variation.
- Modify component exchange pricing and terms to minimize the variations.

The following is an example of using the process to review a dealer 793C rear wheel group flat rate. The analysis is for 51 component rebuilds over a 6 month period.

Work Order Data Summary Spreadsheet

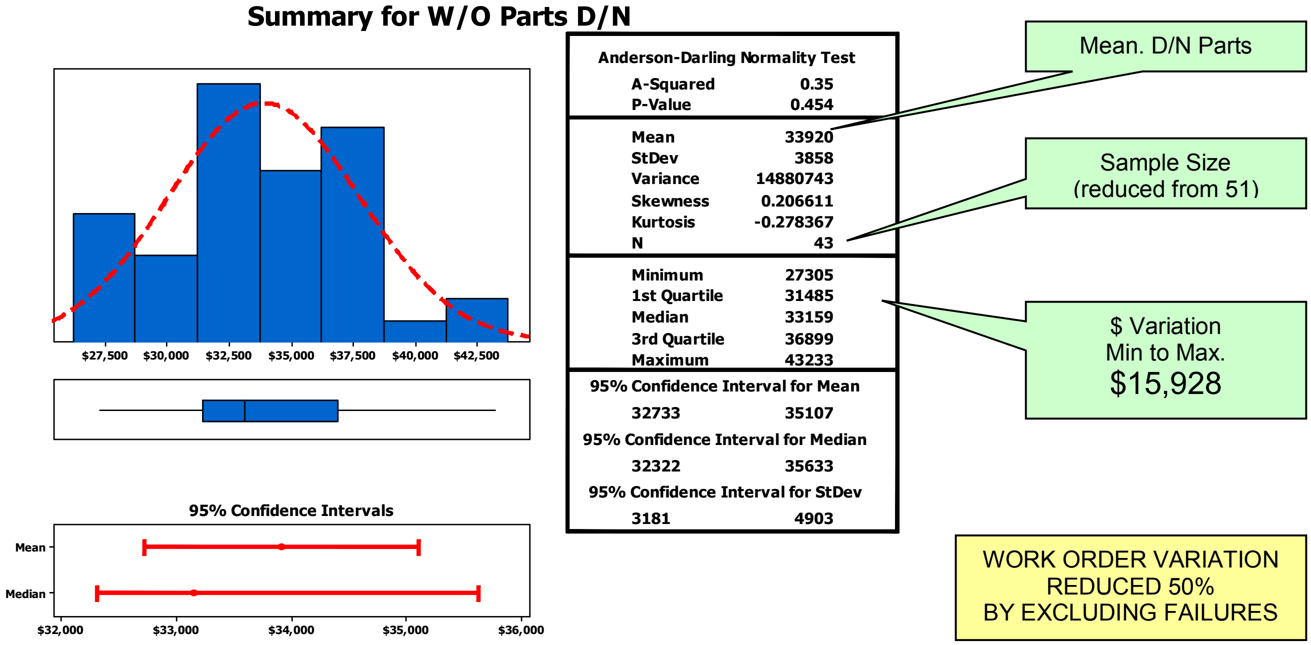
	W/O Parts D/N	Labor Hours	Prior Rebuilds	Customer	Failure?
1	$37,618	37	2	A	N
2	$36,512	39	2	A	N
3	$38,261	41	2	A	N
4	$38,503	36	2	B	N
5	$42,026	44	3	B	N
6	$28,216	47	1	B	N
7	$30,761	46	1	B	N

Minitab Histogram Graph of the work order data



Note: The average D/N (dealer net) parts for all the work orders was $36,422. The median is normally higher or lower than the average if the curve is skewed and not a symmetrical bell curve. In this case, the median D/N is lower, so it is better to use the average price point and not the median as a pricing target.

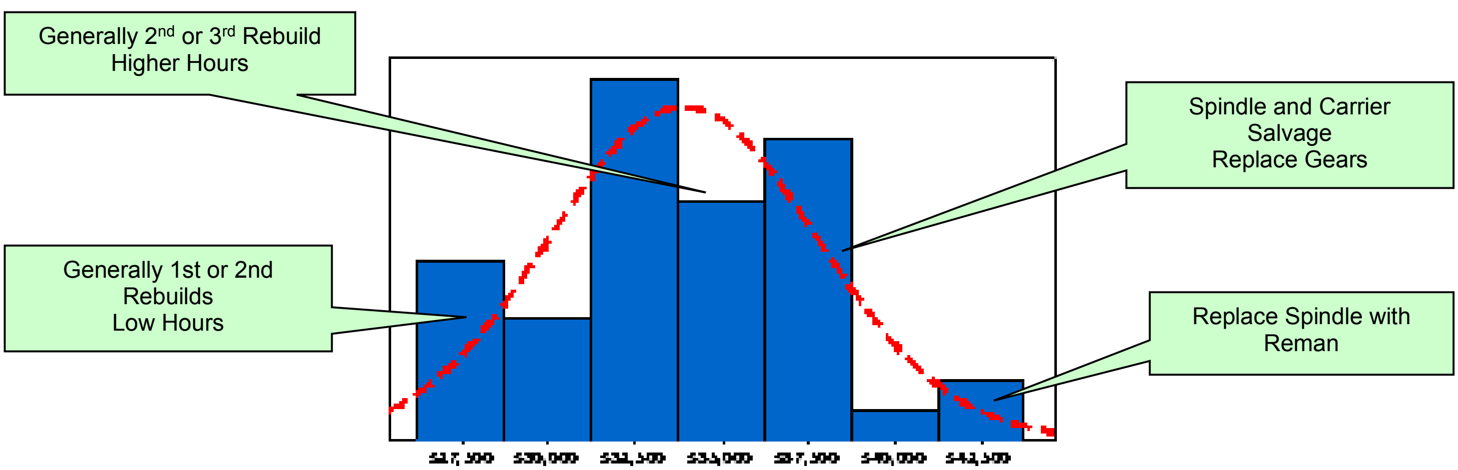
It was determined that the data contained rebuilds for a number of components that were in a failed condition when they were received in the shop. The data was then sorted without the failures included. The variance between the minimum and maximum cost was reduced by 50%, but was still excessive for pricing the component rebuild flat rate. The median rebuild cost was also reduced.



Note: Average D/N parts for the work orders with failure work orders excluded was $33,920. Note that the median is still lower than the average due to the skewed data curve.

The next step in the process was to review the work orders, especially high-dollar work orders, to determine cost drivers.

Cost Drivers



The analysis came to the following conclusions:

- No significant difference in rebuild costs was identified from one customer to the next.
- The major rebuild cost drivers were determined as failures and the aging machine fleets.

The following actions were taken:

- Flat rate for the 793C Final Drive rebuild was defined as a "before-failure" flat rate.
- After-failure rebuilds included a core charge if the spindle, carriers, or hub needed to be replaced.
- Since territory machine fleets were aging, the flat rate would be based on the 3rd quartile D/N cost of $36,899 instead of the average of $33,920. This action was to adjust for future years expecting higher hour component rebuilds.
- Average labor was 49 hours for a cost after direct of $3400.
- Total D/N parts and labor set at $42,299 flat rate.
- There were no product updates affecting Wheel Group rebuilds.
- Crosscheck Reman price: Reman Wheel Group with full core credit D/N $49,959.

Requirements for using this or a similar analysis process:

- Work order records with sufficient detail to identify the needed data and information.
- A component life tracking system to track the previous rebuilds and total hours. Caterpillar provides the Major Component Tracking System free to dealers: MCTS.cat.com
- Sufficient machine population and rebuild volume for an adequate sample size of work orders to review. Fewer than 10 rebuilds on a particular component will lower data confidence.

Analysis Tips

- Understand the difference between "average" (mean) and "median". Average is simply the sum of all the rebuild costs divided by the number of rebuilds. The median is the cost point at which the number of rebuilds below that point equals the number above it. Always compute both the average and the median for comparison.
- Many dealers take the simple approach in averaging a number of rebuilds and set a price at that average. This method does not explain or take into account the variation in rebuild costs from one work order to another. Understanding the variation leads to better decisions.
- There is no rule that says you need to price at the average or the median. In the example above, the price was set at the 3rd quartile (75th percentile) price. This means that ¾ of the rebuilds performed cost less than the price and ¼ of them cost more.
- Crosscheck your flat rate price against Caterpillar Reman. You may also want to compare to "full core credit" versus "partial core credit" Reman prices.
- Make sure that the work order data reflects current parts pricing and not out of date pricing. Also build in forecast part pricing changes for the upcoming year.
- Be sure to incorporate product updates due to design changes. For most components, design changes may not be a cost driver. However, be sure to review changes, especially for engine rebuilds.

4.0 Benefits

- Flat rates are adjusted with changes in costs and market conditions.
- There is a consistent process and data behind the price changes actions.
- CRC financial performance is maintained.

5.0 Resources Required

- Management time to collect data and perform analysis.
- Process can make a good 6Sigma project.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice Process is combined from the flat rate management procedures at several dealers.

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